

Building Blocks of the Web

Lesson Guide: Creating Your First React Student Dashboard

1. Introduction: How Websites Are Built

Have you ever wondered how your school website displays dynamic information like report cards, announcements, or live cricket scores? Modern web applications aren't built as single rigid pages; instead, they are created using reusable building blocks called **Components**.

 **The Lego Analogy:** Think of web components like Lego bricks! Just as you stack individual Lego bricks together to build a castle or a spaceship, developers combine small React components to build full web applications.

2. What is React & JSX?

React is a powerful frontend library that helps developers build interactive user interfaces using reusable component blocks.

JSX (JavaScript XML): React uses JSX, a syntax extension that looks very similar to HTML. JSX makes it easy to write and structure components right inside your JavaScript code — acting like a label or name tag for each block!

3. Functional Components & Props

Functional Components are JavaScript functions that return JSX to display UI elements on the screen.

To make components dynamic, we pass information to them using **Props** (short for properties).

 **The Tiffin Box Analogy:** Props are like school tiffin boxes passed between friends. The box (Component) stays the same, but the food inside (Props data) can change every day!

```
function StudentCard(props) {  
  return (  
    <div className="card">  
      <h2>{props.name}</h2>  
      <p>Grade: {props.grade}</p>  
      <p>House: {props.house}</p>  
    </div>  
  );  
}
```

```
}
```

4. Real-World Applications of React in Schools

React components power many digital systems you interact with daily at school:

Application	How React Is Used
1. Digital Notice Boards	Displays live school announcements and event updates dynamically on hallway screens.
2. Online Report Cards	Allows students and parents to securely view grades, marks, and attendance records anytime.
3. Cricket Scoreboards	Shows real-time match scores, overs, and player statistics during school tournaments.
4. House Leaderboards	Tracks and renders points live for school houses like <i>Agni</i> , <i>Trishul</i> , <i>Prithvi</i> & <i>Jal</i> .

Student Practice & Review Questions

Test your knowledge on React, Components, JSX, and Props!

Q1. What is a React Component best compared to?

- (a) A video game controller
- (b) A Lego building brick
- (c) A complex math formula
- (d) A computer hardware cable

Q2. What does JSX look like in React development?

- (a) HTML
- (b) Pure Python
- (c) Musical notation
- (d) SQL Database queries

Q3. What are "Props" used for in React components?

- (a) Holding up physical computer screens
- (b) Passing information or dynamic data into components
- (c) Playing background music on web pages
- (d) Styling fonts with colors

Q4. Which of the following best describes Functional Components in React?

- (a) Math equations used to calculate school budgets
- (b) Reusable building blocks that describe what should appear on screen
- (c) Hardware components inside a tablet
- (d) Daily school bell timetables

Q5. How would you pass the name "Aarav" into a `StudentCard` component as a prop named `name`?

- (a) `<StudentCard name="Aarav" />`
- (b) `<StudentCard props.name = "Aarav">`
- (c) `StudentCard("Aarav")`
- (d) `<StudentCard -> Aarav>`



Home Coding Challenge!

Build your own customized **Student Dashboard** using CodeSandbox or StackBlitz!

- Create a component that displays your Name, Grade, and School House.
- Try adding a custom prop to display your favorite cricket match score or subject grade.
- Change prop values to see your dashboard automatically update!